

Aymen Rahmani

AI Engineering Student – LLMs, RAG & Agents

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Seeking a 6-month end-of-studies internship (PFE / Master's thesis) from February 2027 – LLMs, RAG & agents

Summary

Final-year engineering student specializing in **LLMs, RAG and agents**. Built a hybrid RAG assistant with a **rigorous evaluation harness** (Recall@5 0.94, MRR 0.95), a multimodal triage agent with guardrails, and a release-gate agent exposed over **MCP**. Two ML/CV internships shipped to production (FastAPI, ONNX).

Education

ESPRIT School of Engineering Sep 2024 – Jul 2027 (expected)
Engineering Degree (Diplôme d'Ingénieur, Master's level), Software & Data Science Engineering Ariana, Tunisia
Relevant coursework: Generative AI & NLP, Deep Learning, Machine Learning, MLOps, Probability & Statistics
Higher Institute of Computer Science (ISI) – Bachelor's Degree (Licence) in Embedded Systems Sep 2021 – Jul 2024

Experience

Computer Vision Intern Jun 2026 – Aug 2026
Supra Tech & Services Tunisia

- PCB defect inspection (**RT-DETR-L + PatchCore**): **77.8% mAP50:95 (+24 pts)** with EIoU + NWD losses.
- Real-time inference at 30 FPS (**ONNX, FastAPI**) with Eigen-CAM explanations for operators.

Machine Learning Intern Jul 2025 – Aug 2025
Web Media International Tunisia

- Email-risk classifier (**XGBoost**, cross-validation): **91% F1**, 90% accuracy on large-scale text data.
- Deployed as a **FastAPI** microservice integrated with a Spring Boot / MySQL backend.

Projects

Regulatory RAG Assistant (GDPR, AI Act, NIS2, DORA) – personal project 2026
Python, sentence-transformers, BM25, ChromaDB, cross-encoder, Azure AI Search, Azure OpenAI, Terraform, pytest

- Hybrid dense + BM25 retrieval** (RRF fusion), **cross-encoder reranking**, article-level chunking and **cited** answers.
- Own **evaluation harness** (50 questions): **Recall@5 0.73 → 0.94, MRR 0.72 → 0.95**, faithfulness 1.00 (LLM-as-a-judge), experiment log.
- Cloud version on **Azure AI Search** with keyless auth (Entra ID), infrastructure as code (**Terraform**), pytest unit tests.

CrisisLink – Multimodal Disaster Triage with LLM Agents (team of 6, in progress) Sep 2026 – now
LLM agents, LangChain, RAG, vision-language models, speech-to-text, n8n, vector database, Docker

- Own the **master triage agent** turning citizen photos and voice notes into P1–P4 rescue tickets.
- Fuse vision, speech and RAG outputs via **typed JSON**; **safety guardrails** (CoT grading), human escalation on conflicts, versioned prompts.

Autonomous Multi-Agent DevOps/MLOps Platform (team of 6, in progress) Sep 2026 – now
LangGraph, Kafka, gRPC/Protobuf, PostgreSQL, Qdrant, SonarQube, Semgrep, Trivy, Gitleaks, GitHub Actions, MCP, Docker

- Own the **Security & Quality agent**: tests, coverage, lint, CVEs (Trivy), secrets (Gitleaks) and SonarQube merged into one **ALLOW / WARN / BLOCK release verdict**.
- Public prototype wired into **CI (GitHub Actions)** and exposed as an **MCP server**; Kafka/gRPC platform where 6 agents fix incidents with human approval.

DriftScope – Data-Drift MCP Server – personal project 2026
Python, MCP, Polars, SciPy, NumPy

- Data-monitoring** tools for agents: **Kolmogorov-Smirnov** test and **PSI** (quantile bins, smoothing), drift policy and synthetic data generator.

Certifications & Activities

NVIDIA DLI (5 certificates, 2025–26): deep learning, transformer-based NLP, LLM app development, prompt engineering, LLM evaluation

AWS Certified ML Engineer – Associate (in progress, exam Dec 2026) | **MASSAI** AI Summer School (ESPRIT, Jun 2025, 30 h)

Hackathons (Apr 2026): participated in “AI for Industrial Zones” (AFI & ESPRIT) and H12 Innovation 3.0 “AI Healing Gabès” (ESPRIT)

Technical Skills

LLMs & agents: LangChain, LangGraph, hybrid RAG (BM25 + dense, reranking), multi-agent systems, MCP, prompt engineering, Ollama, n8n
LLM evaluation: Recall@k, MRR, faithfulness, LLM-as-a-judge, guardrails, prompt test sets
ML & data: Python, PyTorch, Transformers, Scikit-learn, XGBoost, Pandas, SQL, PostgreSQL, Qdrant, ChromaDB
Deployment & cloud: FastAPI, Docker, CI/CD (GitHub Actions), MLflow, Azure (AI Search, OpenAI), Terraform, Kafka, gRPC
Other languages: C++, TypeScript
Languages: Arabic (native), **English (C1, certified)**, French (B2), German (B1–B2)